

INTERNATIONAL A-LEVEL PHYSICS

PH04

Unit 4 Energy and Energy resources

Mark scheme

June 2025

Version: 0.1 Pre-Standardisation



Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from www.oxfordaqa.com

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Additional comments/Guidelines	Mark	AO
01.1	Idea that: it allows for long duration of (close together) recordings OR the temperature change occurs too rapidly to measure this change using a thermometer ✓	Do not allow idea of convenience of placing sensor in test tube	1	AO4

Question	Answers	Additional comments/Guidelines	Mark	AO
01.2	Idea that at this temperature latent heat is released (as the liquid changes to solid) ✓ Idea that latent heat ‘released’ during phase change is equal to energy lost to surroundings so temperature is constant ✓	Alternative for MP2: discussion around kinetic energy of particles being constant as potential energy changes	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
01.3	Any three from: ✓✓✓ • calculation of a $\Delta\theta$ • use of $Q = mc\Delta\theta$ with any $\Delta\theta$ and either c • use of $Q = mL$ • their $Q = mc\Delta\theta$ for liquid added to their mL and their $E = mc\Delta\theta$ for solid Answer that rounds to 2.61×10^4 (J) ✓	eg 80 °C – 58 °C OR 22 °C seen	4	2 × AO2 2 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
01.4	Evidence of use of a tangent drawn on Figure 1 to determine gradient at 300 s ✓ Answer in the range 0.15 to 0.17 (K s ⁻¹) ✓		2	AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
01.5	Any two from: ✓✓ • determines rate of energy transfer from S $E = mc \times 0.055$ • uses their rate of energy transfer = $\frac{kA\Delta\theta}{L}$ to determine their $\Delta\theta$ • uses their reading at 400 s – their $\Delta\theta$ 29 (°C) ✓		3	AO2
Total			12	

Question	Answers	Additional comments/Guidelines	Mark	AO
02.1	Uses $\frac{3}{2} kT$ with T in kelvin ✓ 3.5×10^{-22} (J) ✓	Accept ΔT in MP1	2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.2	Use of $p\Delta V$ ✓ $30(.25)$ (J) ✓		2	1 × AO1 1 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.3	Any two from: ✓✓ • converts temperature to K • uses $N = \frac{pV}{kT}$ OR $N = \frac{N_A pV}{kT}$ • subtracts their N_{up} from their N_{down} OR subtracts V_{up} from V_{down} 7.2×10^{21} ✓		3	1 × AO1 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
02.4	Idea that: external pressure is lower so moving air out of the balloons requires less energy ✓ more energy needed in heating the air as the air starts colder ✓ the total work done on the air as it leaves the balloon is the same if ΔN is the same (since $p\Delta V = \Delta NkT$) ✓		3	AO2
Total			10	

Question	Answers	Additional comments/Guidelines	Mark	AO
03.1	Any two from: ✓✓ - calculates volume of nucleus - calculates mass of nucleus - uses their mass and their volume to calculate a density $2.7 \times 10^{17} \text{ (kg m}^{-3}\text{)} \checkmark$	Allow mass determination using 54 u or $26 m_p$ and $28 m_n$	3	1 × AO1 2 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.2	Any two from: ✓✓ - determines R_0 using data from 03.2 - uses $R = R_0 A^{1/3}$ to determine their A - uses their $A - 26$ to determine their number of neutrons 32 ✓	Allow use of ratios Correct ratio equation Correct A determined	3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.3	Any three from: ✓✓✓ - determines charge on iron nucleus OR charge on alpha particle - uses (change in) electric potential - sets their (change in) electric potential energy equal to initial kinetic energy in some form - determines the mass of an alpha particle from its specific charge. $8.5 \times 10^6 \text{ (m s}^{-1}\text{)} \checkmark$	eg for bullet point 3: $\frac{Q}{4\pi\epsilon_0 r} = \frac{1}{2} \frac{m}{q} v^2 \text{ seen}$	4	1 × AO1 3 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
03.4	$5(.0) \times 10^{-14} \text{ m}$ Idea that closest approach depends on the charge on the target nucleus AND The charge on both nuclei is the same ✓		1	AO1

Total			11	
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Question	Answers	Additional comments/Guidelines	Mark	AO
04.1	Identification of areas for fission and fusion ✓ Description of fission and fusion ✓ Consideration of the difference in binding energy per nucleon at the two locations ✓ Consideration of the impact of the difference in number of nucleons involved, leading to idea that fission releases more energy ✓	Fusion on LHS and fission on RHS Fission: energy released when nucleus splits into smaller nuclei Fusion: energy released when small nuclei combine into larger nuclei BEPN greater on LHS/fusion side Very many more nucleons on RHS/fission side so energy released greater Evidence for MP3 and MP4 may be seen in suitable calculations	4	2 × AO1 2 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.2	Any two from: ✓✓ - determines binding energy for one nucleus - determines their energy released from their binding energy calculations and not including neutrons - converts their energy to J $2.6 \times 10^{-11} \text{ (J)}$ ✓		3	2 × AO2 1 × AO3

Question	Answers	Additional comments/Guidelines	Mark	AO
04.3	Any two from: ✓✓ - calculates the mass of 26 protons in u or kg - calculates the mass of 30 neutrons in u or kg - subtracts mass of iron nucleus (in u or kg) from their total mass in u - converts their difference into MeV 471 (MeV) ✓	Accept use of alternative units Accept range 470 to 480	3	AO3

Total			10	
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Question	Answers	Additional comments/Guidelines	Mark	AO
05.1	Any two from: ✓✓ - converts 481 revolutions to radian - calculates their work done by frictional torque OR deceleration a of wheel - equates their work done to change in kinetic energy using kinetic energy $= \frac{1}{2} I \omega^2$ OR uses $T = I a$ 0.049 (kg m ²) ✓		3	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.2	calculates weight of P ✓ determines angular displacement in radians ✓ uses $T \theta = mg \Delta h$ ✓ 2.6(17) × 10 ⁻³ (N m) ✓	Tolerate 1 sf answer	4	1 × AO3 3 × AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
05.3	Any four from: ✓✓✓✓ • take repeat readings and reject anomalous results • (several repeats and) find half-range and use for uncertainty in angle or height • how to use the uncertainty to determine the percentage uncertainty • combine percentage uncertainties in h and θ • use percentage uncertainty to determine absolute uncertainty		4	3 × AO3 1 × AO4
Total			11	

Question	Answers	Additional comments/Guidelines	Mark	AO														
06.1	The mark scheme gives some guidance as to what statements are expected to be seen in a 1- or 2- mark (L1), 3- or 4- mark (L2) and 5- or 6- mark (L3) answer. Guidance provided in section 3.10 of the ‘Mark Scheme Instructions’ document should be used to assist in marking this question.		6	AO1														
	<table><tr><th>Mark</th><th>Criteria</th></tr><tr><td>6</td><td>All three areas covered in some detail. 6 marks can be awarded even if there is an error and/or parts of one aspect missing.</td></tr><tr><td>5</td><td>A fair attempt to analyse all three areas. If there are several errors or missing parts then 5 marks should be awarded.</td></tr><tr><td>4</td><td>Two areas successfully discussed, or one discussed and two others covered partially. Whilst there will be gaps, there should only be an occasional error.</td></tr><tr><td>3</td><td>One area discussed and one discussed partially, or all three covered partially. There are likely to be several errors and omissions in the discussion.</td></tr><tr><td>2</td><td>Only one area discussed or makes a partial attempt at two areas.</td></tr><tr><td>1</td><td>None of the three areas covered without significant error.</td></tr></table>				Mark	Criteria	6	All three areas covered in some detail. 6 marks can be awarded even if there is an error and/or parts of one aspect missing.	5	A fair attempt to analyse all three areas. If there are several errors or missing parts then 5 marks should be awarded.	4	Two areas successfully discussed, or one discussed and two others covered partially. Whilst there will be gaps, there should only be an occasional error.	3	One area discussed and one discussed partially, or all three covered partially. There are likely to be several errors and omissions in the discussion.	2	Only one area discussed or makes a partial attempt at two areas.	1	None of the three areas covered without significant error.
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	Maximum power output • wind speed • radius of blades • ignore references to density of air • importance of location and reliability/strength of wind • reasons not all energy can be used eg friction in bearings, wind speed does not decrease to zero etc																	
Number and arrangement • size of turbine and available space • idea of wind shadows • consideration of size of turbine related to power output and how close together they can be due to wind shadow																		
Environmental factors • noise • harmful to birds • means land can't be used for anything else • aesthetic impact • wind shadow flicker																		

Question	Answers	Additional comments/Guidelines	Mark	AO
06.2	Determines area of sphere at Earth's orbit OR Multiplies $1.3 (\times 10^3)$ by their area of sphere ✓ 3.8×10^{26} (W) ✓	Condone POT errors in MP1	2	AO2

Question	Answers	Additional comments/Guidelines	Mark	AO
06.3	Idea that: sunlight reflected from clouds/atmosphere/albedo of Earth is not 0 ✓ angle of Sun changes throughout the day/year ✓ not all power from Sun can be used by solar panel / maximum efficiency is about 30% ✓		3	AO3

Total			11	
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Question	Key	Answer	AO
7	A	0.20	AO3
8	D	contain atoms with a small nucleon number.	AO1
9	A	to decrease the number of thermal neutrons	AO1
10	D	$4.5 \times 10^{-9} \text{ kg s}^{-1}$	AO2
11	D	The volume of an individual particle is negligible.	AO1
12	B	$U_1 + Q - U_0$	AO1
13	B	increase decrease	AO4
14	C	$\text{kg s}^{-3} \text{ K}^{-1}$	AO1
15	D	84 kg m^2	AO3
16	C	12 V to 18 V	AO4
17	C	$7.8 \times 10^4 \text{ s}$	AO2
18	A	3.1×10^{-4}	AO3
19	B	4 cm^3	AO2
20	D	Back-up power stations can be started up quickly.	AO1
21	A	0.50 rad s^{-2} clockwise	AO2

Total marks = 15